Downstream Benchmarks of Pareto-Optimal Compressed Models: Qwen2.5-0.5B-Instruct

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Abstract

Two multi-objective evolutionary searches (NSGA-II) compress Qwen2.5-0.5B-Instruct. The first (UBQ-ECSQ) trades calibration loss against encoded storage size, with global quantization and pruning parameters; the second (hardware-aware) trades calibration loss against predicted GPU decode latency, with a per-layer-group choice among bf16 and int4 kernel formats, MLP pruning and sublayer removal. From each Pareto front, the solutions at calibration-loss increases of 0%, 1%, 2%, 5% and 10% are evaluated, with the uncompressed model, on eight standard benchmarks: six zero-shot commonsense and reasoning tasks (PIQA, ARC-Easy, ARC-Challenge, HellaSwag, WinoGrande, BoolQ), 5-shot MMLU and 5-shot GSM8K. UBQ-ECSQ: at 8.0 bits per weight the model is 2.00× smaller with no loss of accuracy; at 6.0–6.7 bits per weight (2.39–2.66× smaller) it keeps 98% of the average accuracy; at 5.0 bits per weight (3.2× smaller) it keeps 88%. Hardware-aware, timed on an RTX 3060: 1.56× faster decoding keeps 98.8% of the accuracy, and 1.98× faster decoding with 3.4× less memory (4.5 bits per weight, almost all int4) keeps 95.1%; removing sublayers costs far more accuracy than it saves time. Hardware-aware, timed on an A100 with a larger search (two and three objectives): the same memory savings keep the same accuracy (98.5% at 1.6× less memory, 94.0% at 3.3×), but decoding is only 1.09–1.17× faster, because on the A100 the weight matrices are a smaller part of a decode step.

1 UBQ-ECSQ compression scheme

All 169 weight matrices of the model are compressed, including the token embedding, which is tied to the output head (494.0M weights, 100% of the parameters). Every matrix uses the same five parameters (genes). For a weight w in output row r, with σ_r and $m_r = \max|w_r|$ taken from the original weights:

1. **Pruning:** $w \to 0$ if $\alpha\,\sigma_r \le w \le \beta\,\sigma_r$, with $\alpha \le 0 \le \beta$.
2. **Uniform quantization (UBQ):** a symmetric, zero-preserving grid of k levels with step $s_r = c\,m_r/\lfloor (k-1)/2 \rfloor$, clipping factor $c \in [0.7, 1]$; pruned weights stay 0.
3. **Entropy-constrained merge (ECSQ):** each non-empty level gets a model-wide centroid; contiguous levels are merged by dynamic programming to minimize $D + \lambda R$ (added squared error plus entropy of level occupancy), keeping a fraction ρ of the levels. The zero level is never merged. $\rho = 1$ means no merge.

Encoded size is the smaller of a dense code ($\lceil \log_2 k_{\text{used}} \rceil$ bits per weight) and a bitmap code (1 bit per weight plus a code per non-zero), plus a 16-bit scale per row and a lookup table per matrix. Sizes are in MB (10^6 bytes); bits per weight include scales and tables.

2 UBQ-ECSQ search

- **Objectives (minimized):** $\Delta\mathcal{L}$, the next-token cross-entropy on the calibration set minus that of the uncompressed model, and the encoded size.
- **Constraint:** $\Delta\mathcal{L} \leq 1.0$ nats.
- **Calibration set:** 8 sequences of 2048 tokens from C4 (train, English), fixed for the whole run. Uncompressed loss $\mathcal{L}_0 = 2.8854$ nats.
- **Genes:** $\log_2 k$ with $k \in [16, 256]$ (range set by a warm-up on the model's error-vs-k curve), $c \in [0.7, 1]$, $\alpha \in [-1, 0]$, $\beta \in [0, 1]$, $\rho \in [0, 1]$.
- **NSGA-II:** population 100, 50 generations, seed 42.

3 UBQ-ECSQ Pareto fronts

Figure 1 shows the final front on the objective of the search, the calibration loss. Figure 2 scores the same 100 solutions on held-out text: perplexity on the full WikiText-2 test split and on 256 windows of C4 validation, with 2048-token windows. The test fronts follow the calibration front: the loss stays near the uncompressed model's down to about 370 MB and rises steeply below 310 MB.

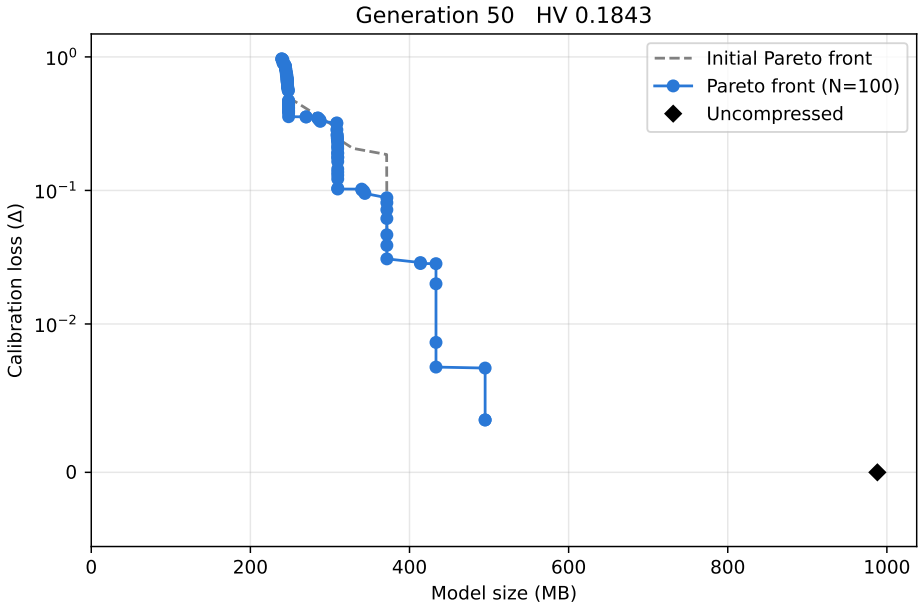
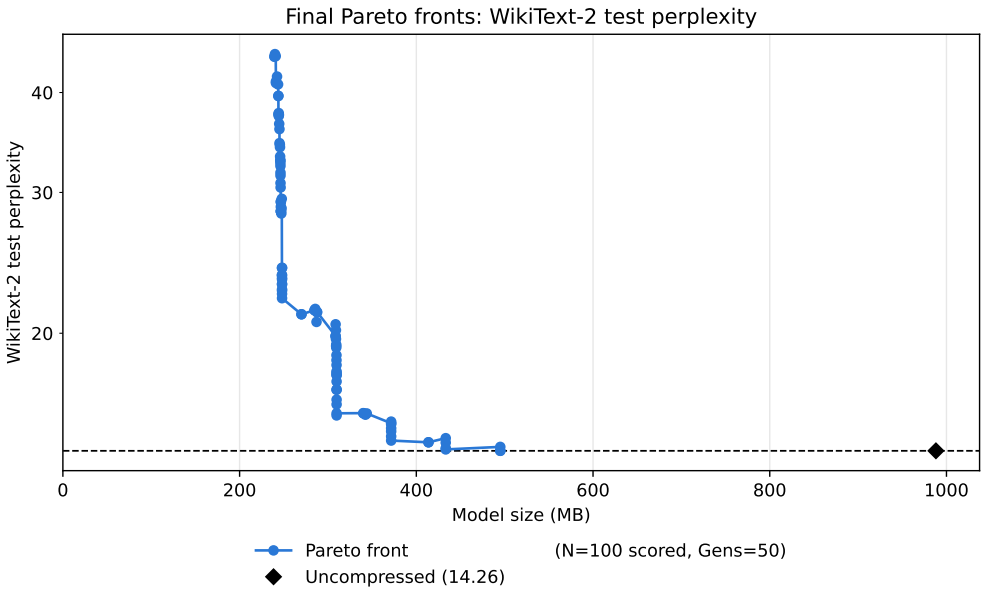
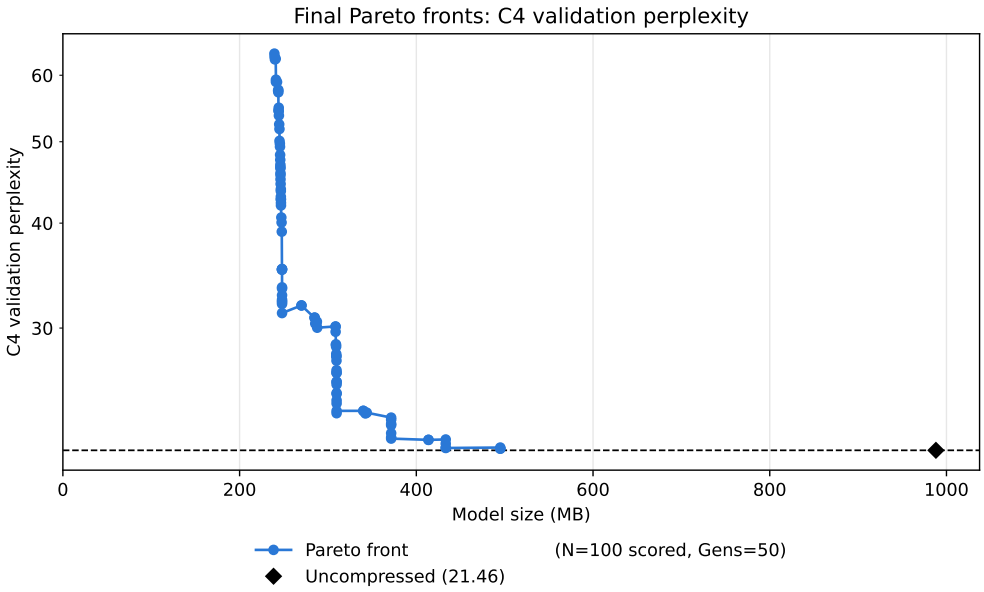


Figure 1: Final Pareto front (100 solutions) of calibration $\Delta\mathcal{L}$ against encoded size, with the front of the initial population and the uncompressed model.



(a) WikiText-2 test



(b) C4 validation

Figure 2: Test perplexity of the final Pareto front against encoded size. The dashed line is the uncompressed model.

4 Selected UBQ-ECSQ models

The calibration error of a solution is $\Delta\mathcal{L}/\mathcal{L}_0$. For each level (0, 1, 2, 5, 10%) the smallest solution within it is selected. No solution has zero error, so the most accurate one (0.12%) stands for the 0% level. Every solution smaller than 309 MB exceeds 10% error, so the 5% and 10% models are nearly the same size but reach it with different settings: Q10 prunes more ($\beta = 0.32$ against 0.02) and clips harder ($c = 0.85$ against 0.93).

Model	k	c	Sparsity (%)	Calib. err. (%)	Size (MB)	Smaller by	Bits/w	Huffman bits/w	WikiText-2 ppl	C4 ppl
Uncompressed (bf16)	–	–	0.0	0.00	988	1.00×	16.00	–	14.26	21.46
Q0 (0%)	164	0.995	2.2	0.12	495	2.00×	8.02	6.47	14.26	21.55
Q1 (1%)	65	0.934	5.3	1.00	414	2.39×	6.70	5.23	14.61	22.08
Q2 (2%)	52	0.949	6.8	1.61	372	2.66×	6.01	4.84	15.08	22.49
Q5 (5%)	29	0.928	11.8	4.75	310	3.19×	5.01	4.05	17.00	25.05
Q10 (10%)	28	0.849	19.2	9.87	309	3.20×	5.00	3.98	20.52	29.70

Table 1: Selected solutions. Sparsity counts pruned weights and weights rounded to zero. Huffman bits/w is the storage with Huffman-coded codes (reported only, not optimized). None of the five merges levels ($\rho = 1$).

5 Benchmark protocol

All models are evaluated with the same harness (lm-evaluation-harness 0.4.13), in bf16, on full evaluation splits, without a chat template, with fixed seeds. Compressed weights are reconstructed in place, so the models differ only in their weight values. The hardware-aware models were evaluated on a second machine together with their own uncompressed reference, which agrees with the first within one standard error on every task; each model is compared with the reference from its own machine. The A100 models of Section 8 reuse the uncompressed reference of the RTX 3060 hardware-aware models with the same settings, without evaluating it again.

Benchmark	What it tests	Split	Questions	Shots	Metric
PIQA	Physical commonsense: the right way to do an everyday task	validation	1,838	0	acc. (norm.)
ARC-Easy	Grade-school science questions	test	2,376	0	acc. (norm.)
ARC-Challenge	Harder grade-school science questions (retrieval-resistant)	test	1,172	0	acc. (norm.)
HellaSwag	Commonsense: the most plausible continuation of an everyday scene	validation	10,042	0	acc. (norm.)
WinoGrande	Commonsense pronoun resolution (fill in the blank, two options)	validation	1,267	0	accuracy
BoolQ	Reading comprehension: yes/no questions about a Wikipedia passage	validation	3,270	0	accuracy
MMLU	Knowledge and reasoning in 57 subjects (STEM, humanities, law, medicine, . . .)	test	14,042	5	accuracy
GSM8K	Multi-step grade-school math word problems, solved in writing	test	1,319	5	exact match

Table 2: Benchmarks. Multiple-choice tasks pick the answer with the highest log-likelihood (acc. (norm.): log-likelihood normalized by answer length); GSM8K generates a solution and compares the extracted final number with the reference.

6 UBQ-ECSQ results

Model	Size (MB)	PIQA	ARC-e	ARC-c	HellaS.	WinoG.	BoolQ	MMLU 5-shot	GSM8K 5-shot	Average
Uncompressed	988	70.4	59.2	33.1	52.4	56.0	67.8	47.0	31.2	52.1
Q0 (0%)	495	70.2	59.9	33.5	52.4	56.6	68.4	47.0	31.7	52.5
Q1 (1%)	414	70.2	59.6	33.7	52.5	55.9	64.3	46.5	27.6	51.3
Q2 (2%)	372	69.5	61.0	32.8	52.0	54.2	66.6	46.1	26.7	51.1
Q5 (5%)	310	69.2	55.2	32.0	50.0	56.4	49.7	45.0	9.7	45.9
Q10 (10%)	309	67.5	56.2	31.0	47.7	57.2	56.1	38.3	12.6	45.8
Std. error		1.1	1.0	1.4	0.5	1.4	0.8	0.4	1.3	

Table 3: Accuracy (%, higher is better), with the unweighted average over the eight benchmarks. Last row: standard error of the uncompressed model's accuracy.

Model	Size (MB)	PIQA	ARC-e	ARC-c	HellaS.	WinoG.	BoolQ	MMLU	GSM8K	Avg. kept (%)
Q0 (0%)	495	−0.2	+0.7	+0.4	−0.1	+0.6	+0.6	0.0	+0.5	100.6
Q1 (1%)	414	−0.2	+0.5	+0.6	0.0	−0.2	−3.5	−0.4	−3.6	98.4
Q2 (2%)	372	−0.9	+1.9	−0.3	−0.5	−1.8	−1.2	−0.8	−4.5	98.0
Q5 (5%)	310	−1.2	−4.0	−1.1	−2.4	+0.3	−18.1	−2.0	−21.5	88.0
Q10 (10%)	309	−2.9	−2.9	−2.1	−4.7	+1.2	−11.8	−8.6	−18.6	87.9

Table 4: Change in accuracy against the uncompressed model (percentage points; negative is a loss). Avg. kept is the compressed model's average accuracy as a percentage of the uncompressed model's.

Observations.

- **8 bits per weight is lossless.** Q0 is 2.00× smaller and every benchmark is within 0.7 points of the uncompressed model, inside the standard errors (100.6% kept).
- **1–2% calibration error keeps 98% of the accuracy.** Q1 and Q2 are 2.39× and 2.66× smaller. The commonsense tasks and MMLU stay within 2 points; GSM8K drops 3.6–4.5 points and BoolQ up to 3.5.
- **At 5.0 bits per weight, about 12% of the average accuracy is lost.** Q5 and Q10 are 3.2× smaller and keep 88%. The commonsense tasks lose at most 5 points; BoolQ breaks (Q5: 49.7%, below the 62.2% of always answering "yes").
- **GSM8K is the most sensitive benchmark:** 31.2% uncompressed, 27–28% at 1–2% error, 10–13% at 5–10% error.
- **Calibration loss does not rank the models downstream.** Q5 has half the calibration error of Q10 and lower test perplexity, and is better on MMLU (45.0 against 38.3), but worse on BoolQ and GSM8K; the two averages are equal. Below about 6 bits per weight, models need a benchmark check before use.

7 Hardware-aware compression on an RTX 3060

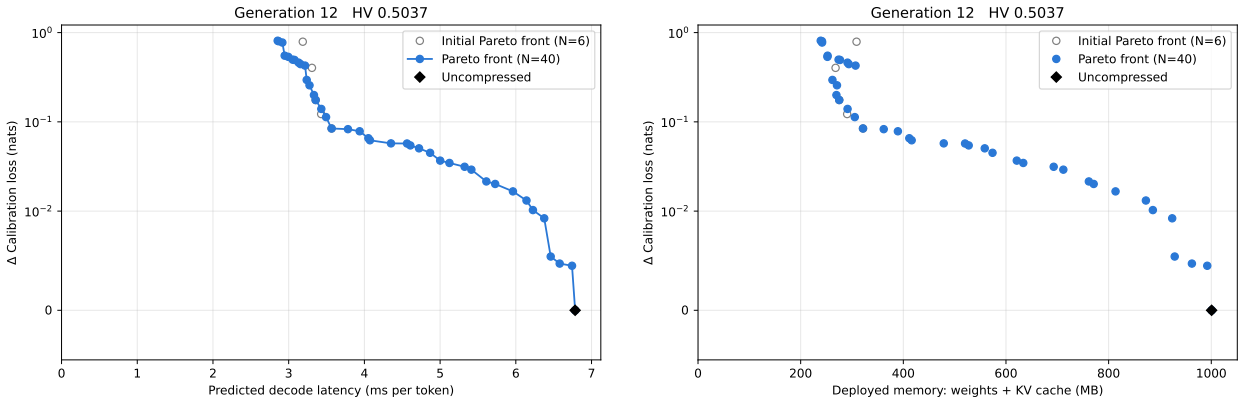
7.1 Scheme and search

The compressed model is the deployed model: every choice maps onto a kernel that runs on the GPU, and the cost is the decode latency and memory of that model, not a storage size.

- **Formats:** bf16, or int4 with an asymmetric bf16 scale and zero per group of 32, 64 or 128 input weights (g32, g64, g128), on PyTorch's int4 matrix-multiplication kernel. The clipping of each int4 group minimizes the activation-weighted error and is solved per group, not searched.
- **Genes (23):** the format of each of four roles (attention q/k/v, attention output, MLP up/gate, MLP down) in each of four depth bands of 6 layers (16 genes); the format of the output head, tied to the token embedding (1); the fraction of MLP neurons kept per band, from $\{1, 0.875, 0.75, 0.625, 0.5\}$, by activation-weighted importance (4); the number of attention and of MLP sublayers removed, 0–6 each, in a fixed order ranked by the loss increase of removing each sublayer alone (2).
- **Objectives (minimized):** $\Delta\mathcal{L}$ on the same calibration set as above ($\mathcal{L}_0 = 2.8854$ nats), and the decode latency of one token (batch 1, 1024 tokens of context), predicted from per-kernel times measured on an RTX 3060 GPU. Memory (weights plus the KV cache of 1024 tokens; 1001 MB uncompressed) is reported, not optimized.
- **Constraint:** $\Delta\mathcal{L} \leq 1.0$ nats. **NSGA-II:** population 40, 12 generations, seed 42.

7.2 Pareto fronts

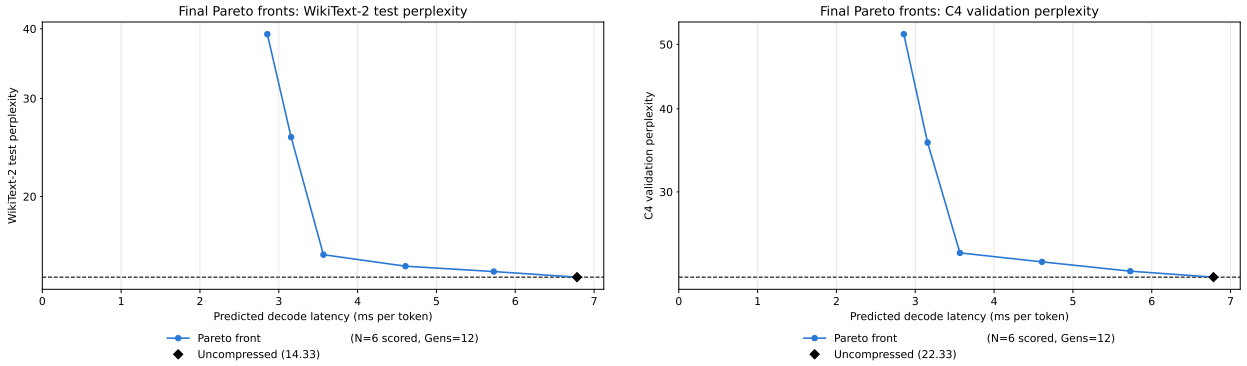
The final front has 40 solutions from 2.85 to 6.78 ms and 239 to 1001 MB (Figure 3). With formats alone, $\Delta\mathcal{L}$ stays at or below 0.18 nats (6%) down to 3.36 ms, the model with every matrix in int4 g128; every faster solution prunes MLP neurons or removes sublayers, and $\Delta\mathcal{L}$ rises to 0.20–0.81 nats. Five solutions spread along the front and the uncompressed model were scored on 32 windows each of WikiText-2 test and C4 validation (Figure 4).



(a) Against predicted decode latency (objective)

(b) Against deployed memory (reported)

Figure 3: Final Pareto front (40 solutions) of calibration $\Delta\mathcal{L}$, with the front of the initial population and the uncompressed model.



(a) WikiText-2 test

(b) C4 validation

Figure 4: Test perplexity of five solutions of the final front and the uncompressed model against predicted decode latency.

7.3 Selected models

For each level (0, 1, 2, 5, 10%) the fastest solution within it is selected (H0–H10). As above, no solution has zero error, so the most accurate one (0.16%) stands for the 0% level.

Model	Calib. err. (%)	Decode (ms)	Faster by	Memory (MB)	Smaller by	Bits/w	int4 (%)	Formats q/k/v out	up	down	head	Removed attn./MLP
Uncompressed	0.00	6.78	1.00×	1001	1.00×	16.00	0	BBBB	BBBB	BBBB	B	0/0
H0 (0%)	0.16	6.74	1.01×	992	1.01×	15.86	1	B3BB	BBBB	BBBB	B	0/0
H1 (1%)	0.75	5.61	1.21×	761	1.31×	12.13	35	BBBB	633B	B36B	B	0/0
H2 (2%)	1.99	4.35	1.56×	479	2.09×	7.55	77	33B3	B333	33B3	3	0/0
H5 (5%)	4.84	3.43	1.98×	291	3.43×	4.51	99	3B38	3386	8888	8	0/0
H10 (10%)	8.90	3.28	2.07×	271	3.70×	4.19	97	8836	6888	8888	8	1/1

Table 5: Selected hardware-aware solutions. Formats: one character per depth band (bands 1–4) for attention q/k/v, attention output, MLP up/gate and MLP down, then the output head; B = bf16, 3/6/8 = int4 with groups of 32/64/128. Bits/w counts weights only; memory includes the 13 MB KV cache. None prunes MLP neurons.

7.4 Results

Model	Decode (ms)	PIQA	ARC-e	ARC-c	HellaS.	WinoG.	BoolQ	MMLU 5-shot	GSM8K 5-shot	Average
Uncompressed	6.78	70.6	59.2	33.0	52.4	55.2	67.7	47.0	30.3	51.9
H0 (0%)	6.74	70.7	59.6	33.4	52.0	55.6	66.9	46.5	31.1	52.0
H1 (1%)	5.61	70.1	60.0	32.7	51.4	56.0	67.6	46.7	27.9	51.5
H2 (2%)	4.35	69.7	59.7	32.8	51.0	56.7	66.1	45.9	28.5	51.3
H5 (5%)	3.43	69.0	58.0	31.5	49.4	55.0	66.5	44.8	20.5	49.4
H10 (10%)	3.28	67.3	49.7	29.0	46.4	52.4	63.8	37.6	4.9	43.9
Std. error		1.1	1.0	1.4	0.5	1.4	0.8	0.4	1.3	

Table 6: Accuracy (%, higher is better) of the hardware-aware models, with the unweighted average.

Model	Decode (ms)	PIQA	ARC-e	ARC-c	HellaS.	WinoG.	BoolQ	MMLU	GSM8K	Avg. kept (%)
H0 (0%)	6.74	+0.2	+0.4	+0.3	−0.4	+0.4	−0.9	−0.5	+0.8	100.1
H1 (1%)	5.61	−0.5	+0.8	−0.3	−0.9	+0.9	−0.1	−0.3	−2.4	99.3
H2 (2%)	4.35	−0.8	+0.5	−0.3	−1.4	+1.5	−1.6	−1.1	−1.8	98.8
H5 (5%)	3.43	−1.5	−1.1	−1.5	−2.9	−0.2	−1.2	−2.2	−9.8	95.1
H10 (10%)	3.28	−3.3	−9.5	−4.0	−5.9	−2.8	−3.9	−9.4	−25.4	84.6

Table 7: Change in accuracy against the uncompressed model (percentage points; negative is a loss).

Observations.

- **The 0% level brings no speed-up.** H0 changes one band of one role to int4 (1% of the weights) and is 1.01× faster; any larger change costs measurable calibration loss.
- **1–2% calibration error: up to 1.56× faster at 98.8% accuracy.** H1 and H2 stay within 1.6 points on every benchmark except GSM8K (−2.4 and −1.8).
- **Almost-all-int4 is the efficient point.** H5 is 1.98× faster with 3.43× less memory and keeps 95.1%: the commonsense tasks and MMLU lose at most 2.9 points, GSM8K 9.8.
- **Removing sublayers does not pay.** H10 removes one attention and one MLP sublayer for 0.15 ms (4%) over H5 and loses another 5.5 points of average accuracy (84.6% kept); GSM8K falls to 4.9%, ARC-Easy and MMLU by 9.5 and 9.4 points.
- **Calibration loss ranks these models correctly:** accuracy falls monotonically with the level, unlike the UBQ-ECSQ models at 5–10%.
- **Against UBQ-ECSQ:** H5 at 4.5 bits per weight keeps 95.1%, the UBQ-ECSQ models at 5.0 bits per weight 88% (BoolQ 66.5 against 49.7, GSM8K 20.5 against 9.7). The UBQ-ECSQ sizes are entropy-coded storage that must be decoded before use; the hardware-aware models run directly on int4 kernels.

8 Hardware-aware compression on an A100

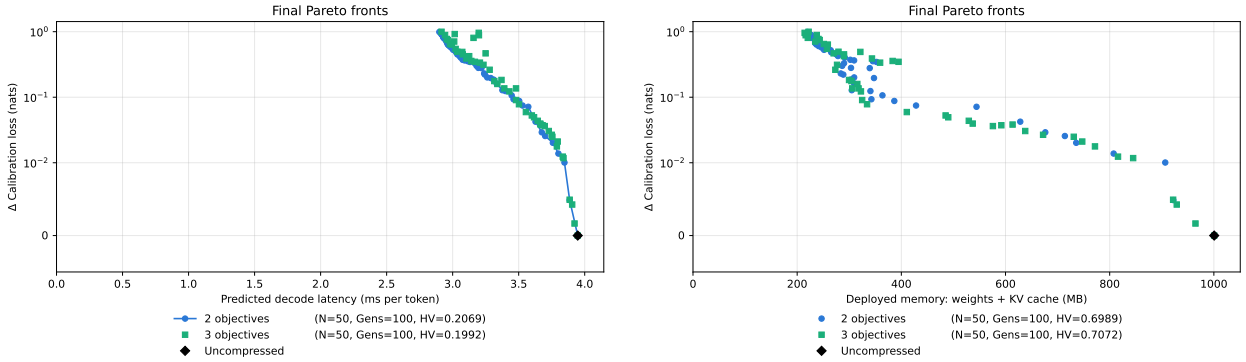
8.1 Search

The scheme, genes (23), calibration set and constraint are those of the previous section. Three things change:

- **GPU:** the kernel times are measured on an NVIDIA A100 80 GB PCIe, with the same workload (batch 1, 1024 tokens of context). The uncompressed model decodes a token in 3.95 ms (6.78 ms on the RTX 3060); memory is again 1001 MB.
- **Search size:** NSGA-II with population 50 for 100 generations (about 5000 evaluated solutions, against 480), seed 42.
- **Two runs:** *2 objectives* minimizes $\Delta\mathcal{L}$ and the decode latency, with memory reported, as before; *3 objectives* also minimizes memory. Both have $\Delta\mathcal{L} \leq 1.0$ nats and memory ≤ 7000 MB as constraints; the memory limit is never reached.

8.2 Pareto fronts

Both final fronts have 50 solutions from 2.9 to 3.95 ms (Figure 5). With the hypervolume of both runs normalized together, the 2-objective front is slightly ahead against latency (0.2069 against 0.1992) and the 3-objective front against memory (0.7072 against 0.6989), where it has more solutions between 400 and 1000 MB.



(a) Against predicted decode latency

(b) Against deployed memory

Figure 5: Final Pareto fronts (50 solutions each) of the 2- and 3-objective A100 searches, with the uncompressed model. HV: hypervolume of each front, with both runs normalized together.

8.3 Selected models

As before, the fastest solution within each level (0, 1, 2, 5, 10%) is selected: A0–A10 from the 2-objective front, B0–B10 from the 3-objective front. No solution has zero error, so the most accurate one stands for the 0% level.

Model	Calib. err. (%)	Decode (ms)	Faster by	Memory (MB)	Smaller by	Bits/w	int4 (%)	MLP kept (%)	Formats q/k/v out	up	down	head	Removed attn./MLP
Uncompressed	0.00	3.95	1.00×	1001	1.00×	16.00	0	100	BBBB	BBBB	BBBB	B	0/0
2 objectives													
A0 (0%)	0.17	3.89	1.02×	922	1.09×	14.73	12	100	BBBB	BBB3	B3BB	B	0/0
A1 (1%)	0.89	3.70	1.07×	714	1.40×	11.36	42	100	3BBB	3333	333B	B	0/0
A2 (2%)	1.47	3.63	1.09×	628	1.59×	9.97	54	100	3BB3	6333	333B	6	0/0
A5 (5%)	4.44	3.38	1.17×	305	3.28×	4.74	100	100	3336	3333	6663	3	1/0
A10 (10%)	9.72	3.19	1.24×	303	3.30×	4.74	99	100	8638	3338	6363	3	4/0
3 objectives													
B0 (0%)	0.06	3.92	1.01×	965	1.04×	15.42	5	100	BBBB	BBBB	BBBB	B	0/0
B1 (1%)	0.93	3.75	1.05×	672	1.49×	10.68	48	100	BBBB	BBBB	3363	B	0/0
B2 (2%)	1.83	3.60	1.10×	485	2.06×	7.65	75	100	3BBB	6366	336B	3	0/0
B5 (5%)	4.73	3.39	1.16×	306	3.27×	4.75	98	100	66B8	3366	3663	6	1/0
B10 (10%)	9.10	3.28	1.20×	273	3.66×	4.24	95	93	3663	6883	3668	8	2/0

Table 8: Selected A100 solutions. Formats as in Table 5: one character per depth band for each role, then the output head; B = bf16, 3/6/8 = int4 with groups of 32/64/128. MLP kept: the fraction of MLP neurons kept over the whole model (B10 keeps 75% in the second band). Memory includes the 13 MB KV cache.

8.4 Results

Model	Decode (ms)	PIQA	ARC-e	ARC-c	HellaS.	WinoG.	BoolQ	MMLU 5-shot	GSM8K 5-shot	Average
Uncompressed	3.95	70.6	59.2	33.0	52.4	55.2	67.7	47.0	30.3	51.9
2 objectives										
A0 (0%)	3.89	70.5	60.9	33.2	52.3	55.4	68.5	46.6	30.5	52.2
A1 (1%)	3.70	70.9	58.8	32.4	51.4	55.6	67.1	46.6	28.1	51.4
A2 (2%)	3.63	70.0	59.8	32.7	50.7	56.7	66.2	46.0	27.1	51.1
A5 (5%)	3.38	68.2	55.4	30.5	49.2	54.6	67.2	45.6	19.6	48.8
A10 (10%)	3.19	67.2	58.1	30.5	46.9	55.7	60.2	41.6	5.8	45.8
3 objectives										
B0 (0%)	3.92	70.3	58.8	33.4	52.2	56.1	67.1	47.1	30.6	52.0
B1 (1%)	3.75	69.9	59.3	32.5	51.5	57.3	67.4	46.2	29.9	51.8
B2 (2%)	3.60	69.6	58.0	31.7	51.0	55.8	65.7	46.1	27.9	50.7
B5 (5%)	3.39	67.9	55.0	31.1	49.1	55.4	66.5	45.1	18.3	48.5
B10 (10%)	3.28	67.4	56.1	29.4	45.4	54.0	62.0	33.3	2.6	43.8
Std. error		1.1	1.0	1.4	0.5	1.4	0.8	0.4	1.3	

Table 9: Accuracy (%, higher is better) of the A100 models, with the unweighted average.

Model	Decode (ms)	PIQA	ARC-e	ARC-c	HellaS.	WinoG.	BoolQ	MMLU	GSM8K	Avg. kept (%)
2 objectives										
A0 (0%)	3.89	−0.1	+1.8	+0.2	−0.1	+0.2	+0.8	−0.3	+0.2	100.6
A1 (1%)	3.70	+0.4	−0.4	−0.6	−1.0	+0.5	−0.6	−0.4	−2.2	99.0
A2 (2%)	3.63	−0.6	+0.6	−0.3	−1.7	+1.5	−1.5	−1.0	−3.3	98.5
A5 (5%)	3.38	−2.3	−3.7	−2.5	−3.2	−0.6	−0.5	−1.4	−10.7	94.0
A10 (10%)	3.19	−3.3	−1.1	−2.5	−5.5	+0.6	−7.5	−5.4	−24.6	88.1
3 objectives										
B0 (0%)	3.92	−0.3	−0.3	+0.3	−0.1	+0.9	−0.6	+0.1	+0.2	100.1
B1 (1%)	3.75	−0.7	+0.2	−0.5	−0.9	+2.1	−0.3	−0.8	−0.4	99.7
B2 (2%)	3.60	−1.0	−1.2	−1.4	−1.4	+0.6	−2.0	−0.9	−2.4	97.7
B5 (5%)	3.39	−2.7	−4.2	−2.0	−3.2	+0.2	−1.2	−1.9	−12.1	93.5
B10 (10%)	3.28	−3.2	−3.1	−3.7	−6.9	−1.2	−5.7	−13.7	−27.7	84.3

Table 10: Change in accuracy of the A100 models against the uncompressed model (percentage points; negative is a loss).

Observations.

- **Accuracy follows the calibration error as on the RTX 3060.** Up to 2% error the models keep 97.7–100.6% of the average accuracy, at 5% 93.5–94.0%, at 10% 84.3–88.1%; in both runs the average falls monotonically with the level.
- **The memory savings carry over; the speed-up does not.** A5 and B5 have the memory and accuracy of H5 (305 against 291 MB; 94.0 and 93.5 against 95.1% kept), but are only

1.16–1.17× faster, against 1.98×. The A100 decodes the uncompressed model in 3.95 ms, against 6.78 ms, and int4 saves less of that time.

- **GSM8K is again the most sensitive benchmark:** 30.3% uncompressed, 27–31% up to 2% error, 18–20% at 5%, 3–6% at 10%.
- **Removing sublayers still does not pay.** On the A100 the 5% models already remove one attention sublayer. A10 removes four for 0.19 ms (6%) over A5 and keeps 88.1% against 94.0%; B10 removes two and prunes a quarter of the MLP neurons of one band, and loses 13.7 points on MMLU and 27.7 on GSM8K.
- **2 against 3 objectives:** at 1–2% error the 3-objective models use less memory (672 against 714 MB, 485 against 628 MB) at the same speed, within 0.05 ms. Up to the 5% level their averages differ from the 2-objective models' by at most 0.4 points, about the standard error of an eight-task average; at 10% B10 is 2.0 points lower.